“SpeekFree”
AI-Based Speech Disorder Screening and Therapy Assistant
Progress Presentation 1

R26-IT-086

Information Technology

SpeakFree

Problem & Its Impact / Gravity

Speech disorders in children and adults can affect communication, education, confidence, and social participation.
Traditional screening often depends on manual clinical observation, which can be time-consuming, subjective, and difficult to access in remote or low-resource areas.
Many patients do not receive early screening or continuous therapy support due to limited access to speech therapists.
Existing digital tools often provide only basic speech practice and lack AI-based screening, severity analysis, articulation analysis, and personalized therapy guidance.

Who is affected / who benefits

- **Children with Speech delays**
- **Speech Therapists**
- **Parents and Caregivers**
- **Schools and clinics**
- **Remote patients with limited access to therapy services**

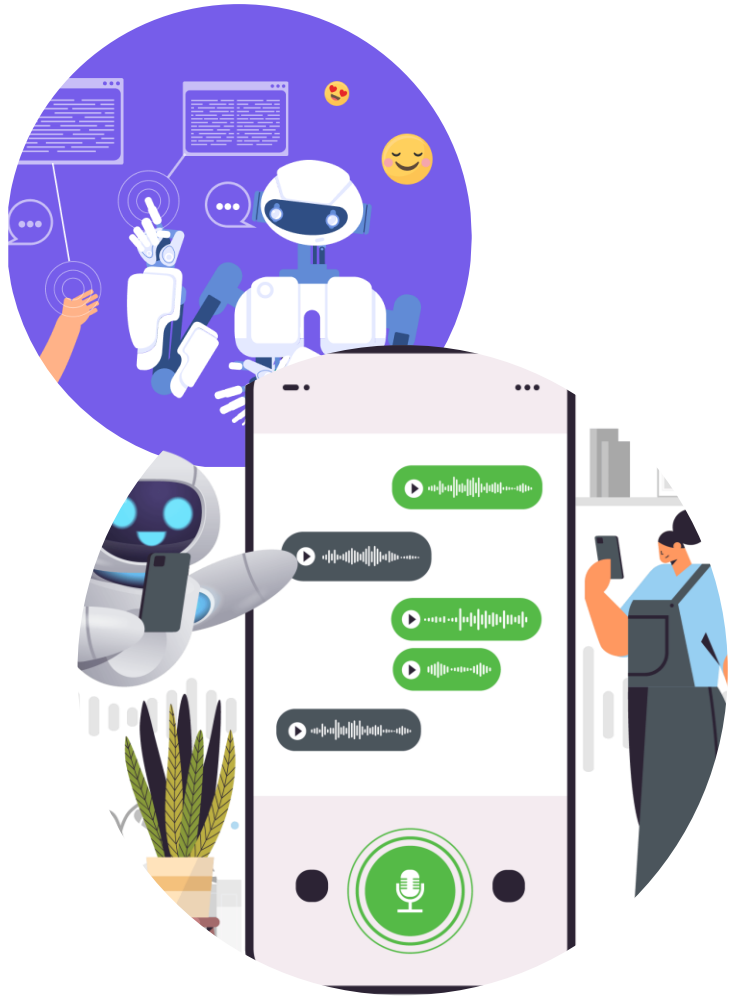


Overall Solution & Design Excellence

SpeakFree is an AI-powered speech disorder screening and therapy platform that supports early screening, articulation analysis, severity prediction, and therapy guidance through a web-based system.

Main components

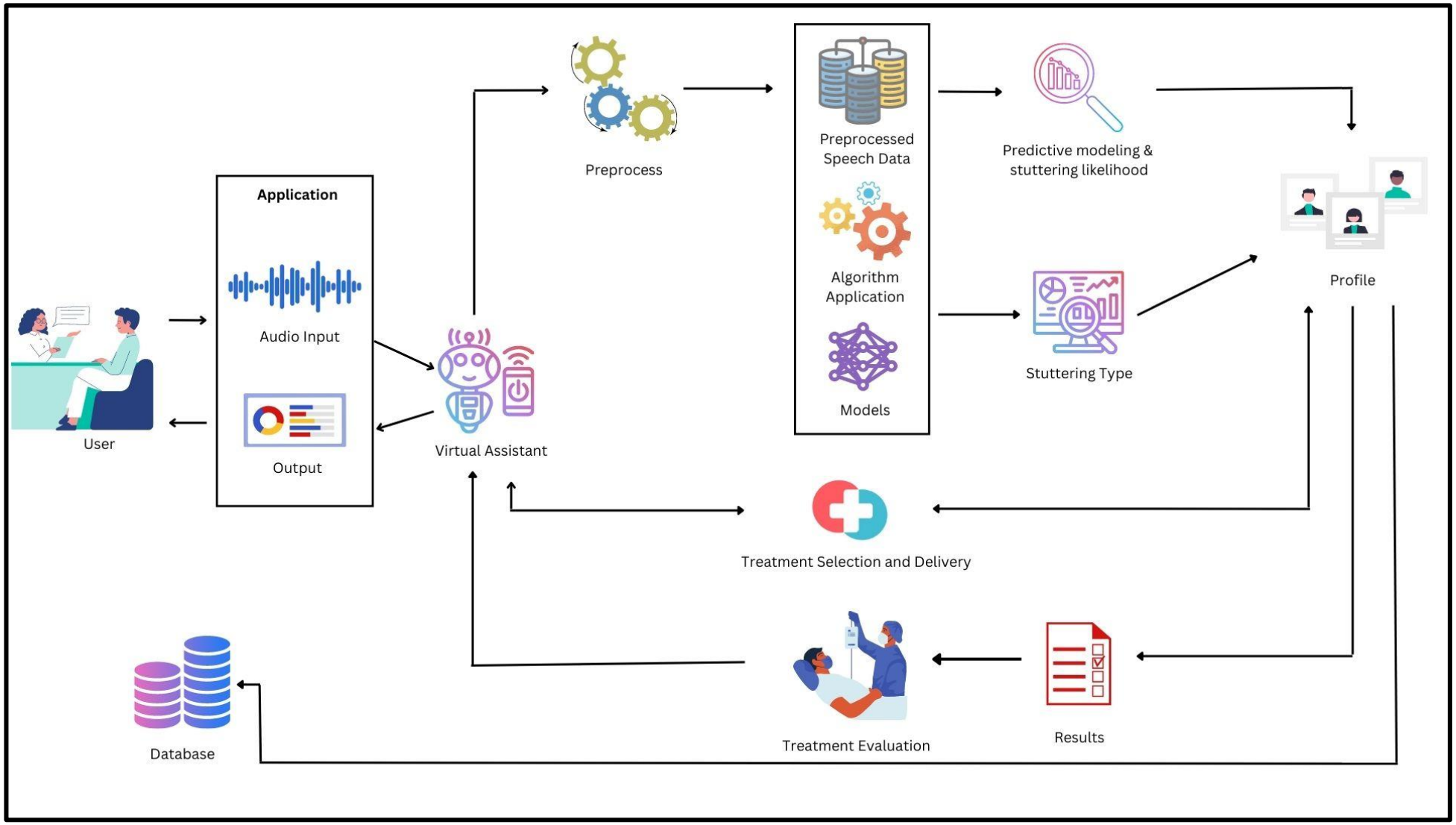
1. **Component 1 – Feature Extraction / Screening Input Processing**
2. **Component 2 – Alignment & Articulation Analytics**
3. **Component 3 – Multi-Disorder Severity Classification & Risk Prediction**
4. **Component 4 – AI Therapy Sessions**



Tools and Technologies

1. **React.js** - Used to build the frontend web application, including assessment screens, dashboards, upload pages, and therapy interfaces.
2. **FastAPI** - Used to build the backend API for handling requests, file uploads, and communication between frontend and AI services.
3. **Python** - Used for backend development and speech processing logic.
4. **Librosa** - Used for extracting audio features such as duration, energy, zero-crossing rate, and tempo.
5. **ASR / Whisper-based Processing** - Used to convert speech into text and generate ASR score outputs.
6. **JSON** - Used to store and transfer structured outputs between components.
7. **VS Code** - Used as the main development environment.
8. **Git and GitHub** - Used for version control, code sharing, and team collaboration.
9. **Database / File Storage** - Used to store patient details, assessment records, audio files, reports, and therapy progress.

SpeakFree System Architecture





AI-BASED SPEECH INPUT FEATURE EXTRACTION AND INITIAL SCREENING MODULE

Speech Processing and Generates a structured **ASR score JSON output** from recorded speech for further analysis and calculations.

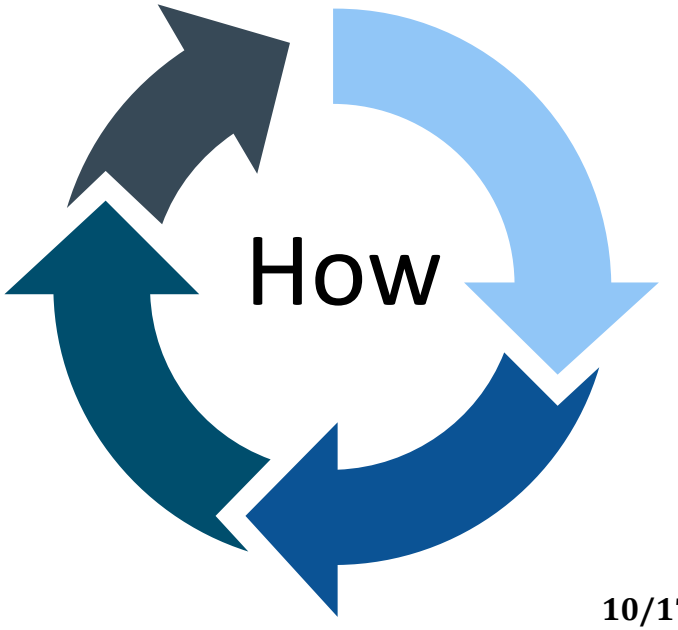
IT22132628 - P.A.S.S. Kusumsiri
Information Technology

HOW THE SOLUTION ADDRESSES THE SUB-PROBLEM / PROTOTYPE

- This component handles the first stage of the screening process by collecting user input such as speech audio, picture description responses, and scenario-based responses.
- The current prototype supports basic audio upload and feature extraction through the backend. Extracted features include duration, energy, zero-crossing rate, tempo, and initial articulation-related indicators.
- The module prepares structured output for later articulation and severity analysis.

User Requirements Addressed by the Solution

- Patient can provide speech input through the web interface.
- System can accept audio files for screening.
- System can display extracted speech features.
- Therapist can review early screening indicators.
- Output can be passed to other AI components.



DESIGN EXCELLENCE

- Designed a modular input processing workflow for speech screening.
- Integrated frontend upload interface with FastAPI backend.
- Structured feature extraction outputs in JSON format.
- Added support for future picture description and scenario description screening tasks.
- Created a foundation for clinically explainable screening results.

CONTRIBUTION

This component is the heart of the system because it converts raw speech input into measurable data used for articulation analysis, severity prediction, and therapy recommendations.



ALIGNMENT & ARTICULATION ANALYTICS FOR SPEECH DISORDER SCREENING

This Component generates articulation and alignment-based speech analysis results, which are passed to Component 3 to classify disorder severity and predict patient risk level.

IT22257086 - Randiw E.Y
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HOW THE SOLUTION ADDRESSES THE SUB-PROBLEM / PROTOTYPE

- This component analyzes how accurately the patient articulates phonemes and words.
  It uses forced alignment to compare spoken audio with expected phonetic patterns.
- The prototype focuses on identifying timing irregularities, phoneme duration deviations, and possible misarticulation patterns.
- The results are converted into interpretable articulation indicators for therapist review.

User Requirements Addressed by the Solution

- Therapist can view articulation-related results.
- System can identify pronunciation timing issues.
- System can generate phoneme-level feedback.
- Patient results can be used for further severity classification.

DESIGN EXCELLENCE / CONTRIBUTION

- Introduces forced-alignment-based articulation analysis.
- Uses interpretable sub-scores such as phoneme duration deviation and articulation precision.
- Adds confidence indicators to reduce unsafe AI decisions.
- Supports explainable clinical review instead of black-box output.
- Provides structured results for Component 3 severity prediction.

MULTI-DISORDER SEVERITY CLASSIFICATION AND RISK PREDICTION



Machine Learning & AI-Based Clinical Speech Analysis

IT21315114 –Tharushan I L S

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HOW THE SOLUTION ADDRESSES THE SUB-PROBLEM / PROTOTYPE

Problem Statement

Many speech screening systems only detect whether a disorder exists, but they do not estimate severity levels or support multi-disorder classification.

This component addresses that limitation by:

- **Predicting severity levels**
- **Supporting multiple disorder types**
- **Providing structured AI-based outputs**

Architecture Flow Diagram

Component 1
Speech Features
↓
Component 2
Articulation Features
↓
Component 3
ML Classification Engine
↓
Severity Prediction
(Level 1–4)
↓
Clinical Report

User Requirements Addressed

- Therapist can view severity predictions
- Early disorder risk identification
- Session-by-session patient monitoring
- Therapy planning support
- Structured clinical reporting

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DESIGN EXCELLENCE / CONTRIBUTION

01

Multi-Disorder AI Architecture

Designed a modular AI pipeline capable of handling:

Stuttering
Dysarthria
Articulation-related disorders

02

Feature Fusion Strategy

Combined:

- Speech fluency features
- Articulation features
- Acoustic indicators

03

Severity Prediction Engine

Implemented:

- Severity scoring
- Confidence scoring
- Risk-based classification

04

Explainable AI Support

Supports:

- Structured clinical metrics
- Transparent prediction outputs
- Future explainability integration

05

Novelty / Innovation

Research Novelty

- Multi-disorder severity prediction
- Unified severity scoring mechanism
- Confidence-aware prediction pipeline
- Foundation for uncertainty estimation
- Bias-aware future model evaluation



AI-BASED ADAPTIVE SPEECH THERAPY RECOMMENDATION COMPONENT

Generates personalized therapy sessions using speech analysis results, severity level, and user progress.

IT22056702 - M.D.I. Perera
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HOW THE SOLUTION ADDRESSES THE SUB-PROBLEM / PROTOTYPE

Many speech screening systems only identify the speech problem or severity level. After that, they do not guide the patient with suitable therapy exercises.

Most therapy exercises are general and not personalized according to the patient's weak sounds, severity level, or progress.

This component solves this by:

- Creating personalized therapy sessions
- Selecting exercises based on weak phonemes
- Changing difficulty according to progress
- Supporting home-based therapy practice

Component 1
Speech input and feature extraction
↓
Component 2
Finds weak sounds / articulation issues
↓
Component 3
Predicts severity level
↓
Component 4
AI therapy recommendation engine
↓
Personalized therapy plan
↓
User practices exercises
↓
Feedback and progress tracking

- Patient can receive suitable therapy exercises
- System can recommend exercises based on weak sounds
- System can provide word-level and sentence-level practice
- System can give repetition-based activities
- System can track therapy progress
- Caregiver or therapist can view progress
- System can recommend therapist review for severe cases

SIMPLE PROTOTYPE EXAMPLE

If the patient has difficulty with the **/s/** sound and the severity level is **moderate**, the system recommends:

- Practice words: **sun, sea, sip, snake, school**
- Practice sentences: **Sara saw seven stars. / The sun shines softly.**
- Repetition count: **5 times each**
- Next review: **After 3 sessions**
- Feedback: **Practice slowly and focus on clear /s/ sound production.**

DESIGN EXCELLENCE / CONTRIBUTION

Adaptive Therapy Architecture

Designed a modular therapy pipeline that receives outputs from previous components and converts them into therapy support. Handles:

- Weak phonemes
- Articulation score
- Severity level
- Confidence score
- User progress

Personalized Recommendation Strategy

Combines:
- Weak sound areas
- Disorder type
- Age group
- Severity level
- Previous therapy Results

to generate suitable therapy exercises for each user.

Progress-Based Difficulty Adjustment

Implements:
- Beginner, intermediate, and advanced levels
- Repetition-based activities
- Word-level practice
- Sentence-level practice
- Next-session recommendations

Difficulty changes according to user improvement.

Explainable Therapy Support

Supports:
- Clear therapy goals
- Simple exercise instructions
- Feedback after practice
- Progress summary for caregivers or therapists
- Therapist referral for severe or uncertain cases

Novelty / Innovation

Research Novelty
- Connects AI screening results with therapy recommendation
- Generates personalized therapy plans instead of general exercises
- Adapts therapy based on user progress • Supports continuous home-based therapy
- Creates a complete screening-to-therapy pathway

COMMERCIALIZATION / BUSINESS POTENTIAL

Customer Persona

- Speech therapy clinics
- Hospitals
- Schools
- Parents and caregivers
- Remote patients



Market Need

- Limited access to speech therapists
- High need for early screening
- Growing demand for digital healthcare tools
- Need for affordable remote therapy support

Commercialization / Business Potential

What Is Unique

- Combines screening, articulation analysis, severity prediction, and therapy sessions in one platform.
- Provides explainable AI outputs for therapist review.
- Supports remote and continuous therapy.
- Designed for both clinical and patient-friendly use.

Cost Recovery

- Subscription for clinics
- School-based licensing
- Therapist dashboard package
- Paid progress reports

Expenses	
Requirement	Cost (Rs)
Domain name	4000.00
Web Hosting/Cloud Server	12000.00
Database/Storage	5000.00
Development Tools	0
Testing and Documentation	3000.00
Miscellaneous Expenses	5000.00
Total Estimated Budget	29000.00

Current Progress

Area	Current Status
Frontend Structure	Completed for all 4 components
Component 1 backend	Basic audio upload and feature extraction available
Component 2	Prototype-level articulation analysis flow
Component 3	Mock/sample severity output flow
Component 4	Therapy session UI and sample recommendation flow
Final integration	Planned for next phase

Risk and Mitigation

Risk	Mitigation
Low-quality speech/audio input	Add audio quality validation
AI prediction errors	Use confidence scores and therapist review
Limited dataset availability	Use public datasets and phased validation
Backend model delay	Use mock API outputs during frontend development
Privacy concerns	Use secure upload handling and minimum patient data storage

Conclusion and Next Steps

- Complete backend integration for all components.
- Improve model accuracy and validation.
- Complete therapy recommendation logic.
- Add report generation.
- Conduct testing with sample speech inputs.
- Prepare for Progress Presentation 2 with stronger implementation evidence.

SpeakFree

